

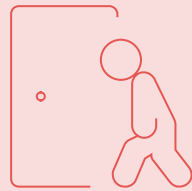
Kartikee

AI-Powered Active
Learning & Therapy
Support

Accessible & Personalized
Pediatric Therapy



The Challenge in Autism Support



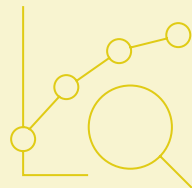
Accessibility

Expensive specialist therapy, limited availability, and travel difficulties.



Engagement

Children lose focus in long sessions; fixed approaches don't work for everyone.



Progress Tracking

Lack of clear session-level insights for parents.



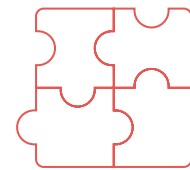
Central Question

How can we make therapy more accessible, engaging, and personalized?

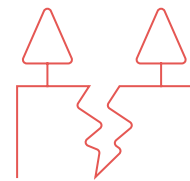
Kartikee bridges the gap in autism support.



Lack
Personalization



Limited
Integration



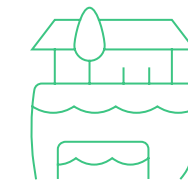
Separated
Experience



Continuous
Engagement



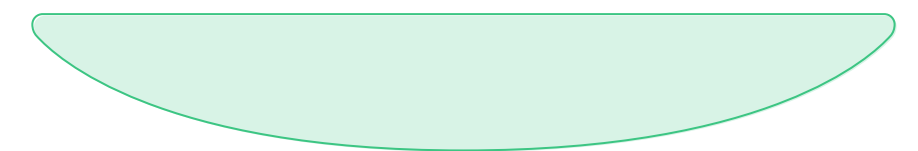
Seamless
Integration



Connected
Experience



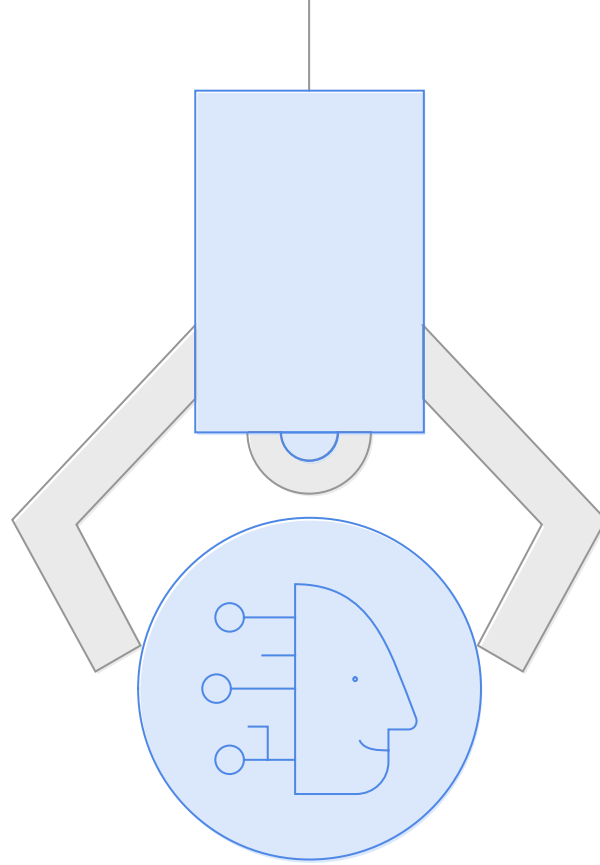
Existing Solutions



Kartikee

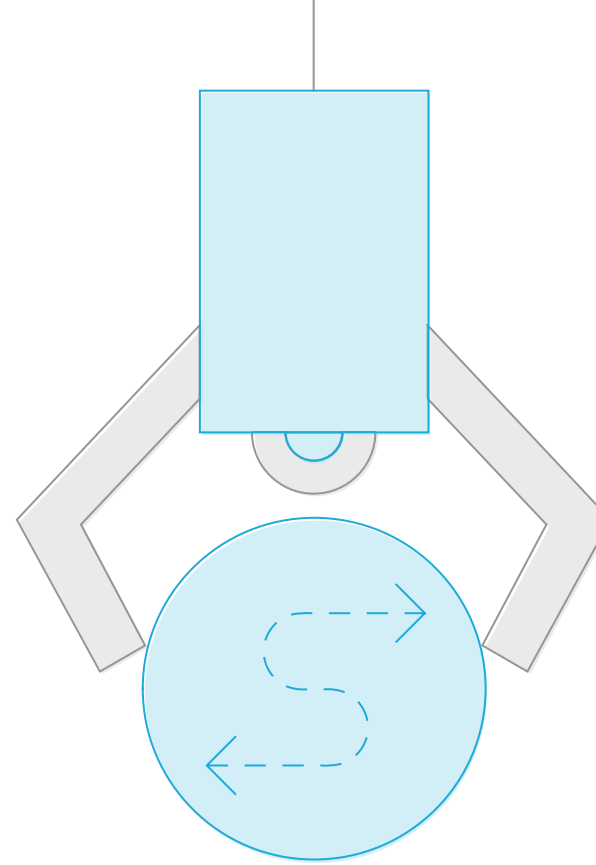
Proposed Solution

A session that respond to observable changes in the childs engagemnet and activity performance



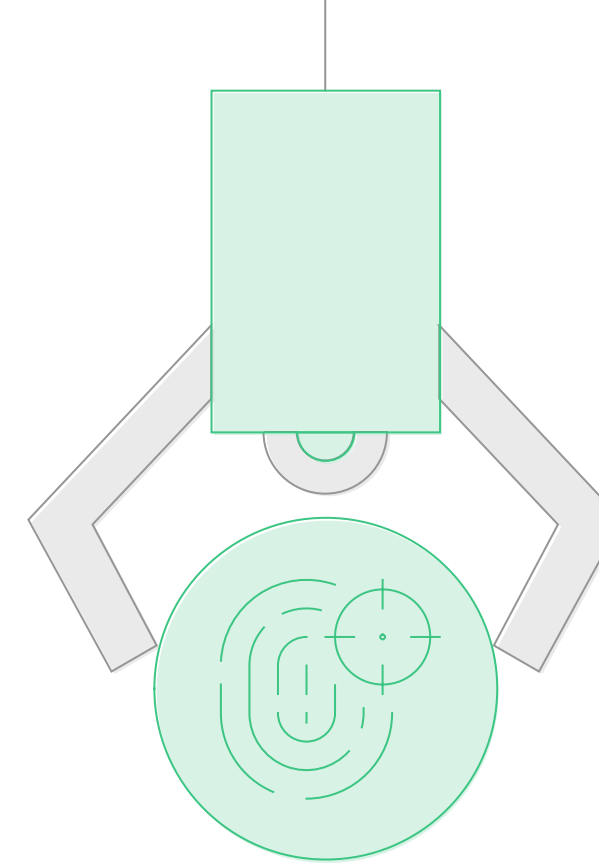
Platform purpose

**An AI-powered
platform designed to
make pediatric
therapy interactive
and personalized.**



The Shift

**Moving from fixed
therapy flows to
sessions that
respond to
observable changes
in engagement and
performance.**

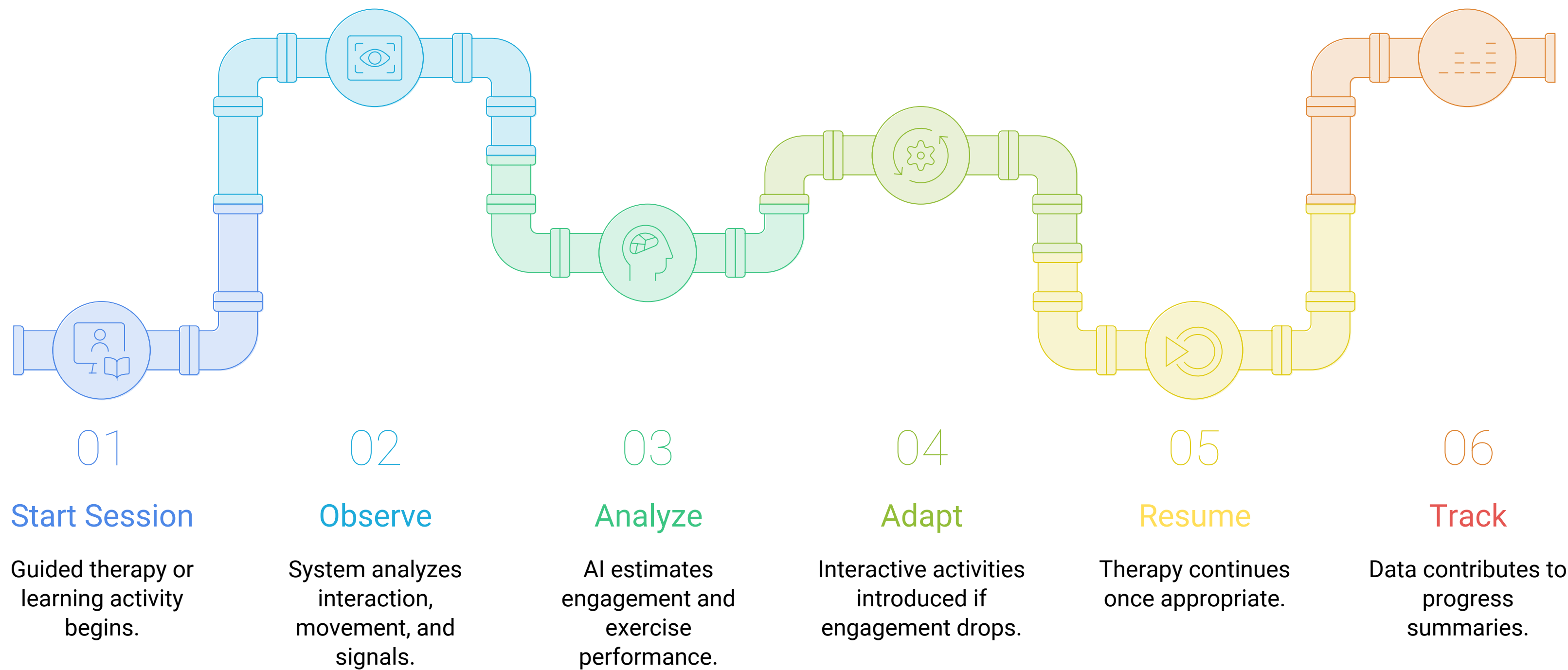


The Principle

**Therapy goals
remain the priority,
while AI provides an
adaptive support
layer.**

Proposed Solution

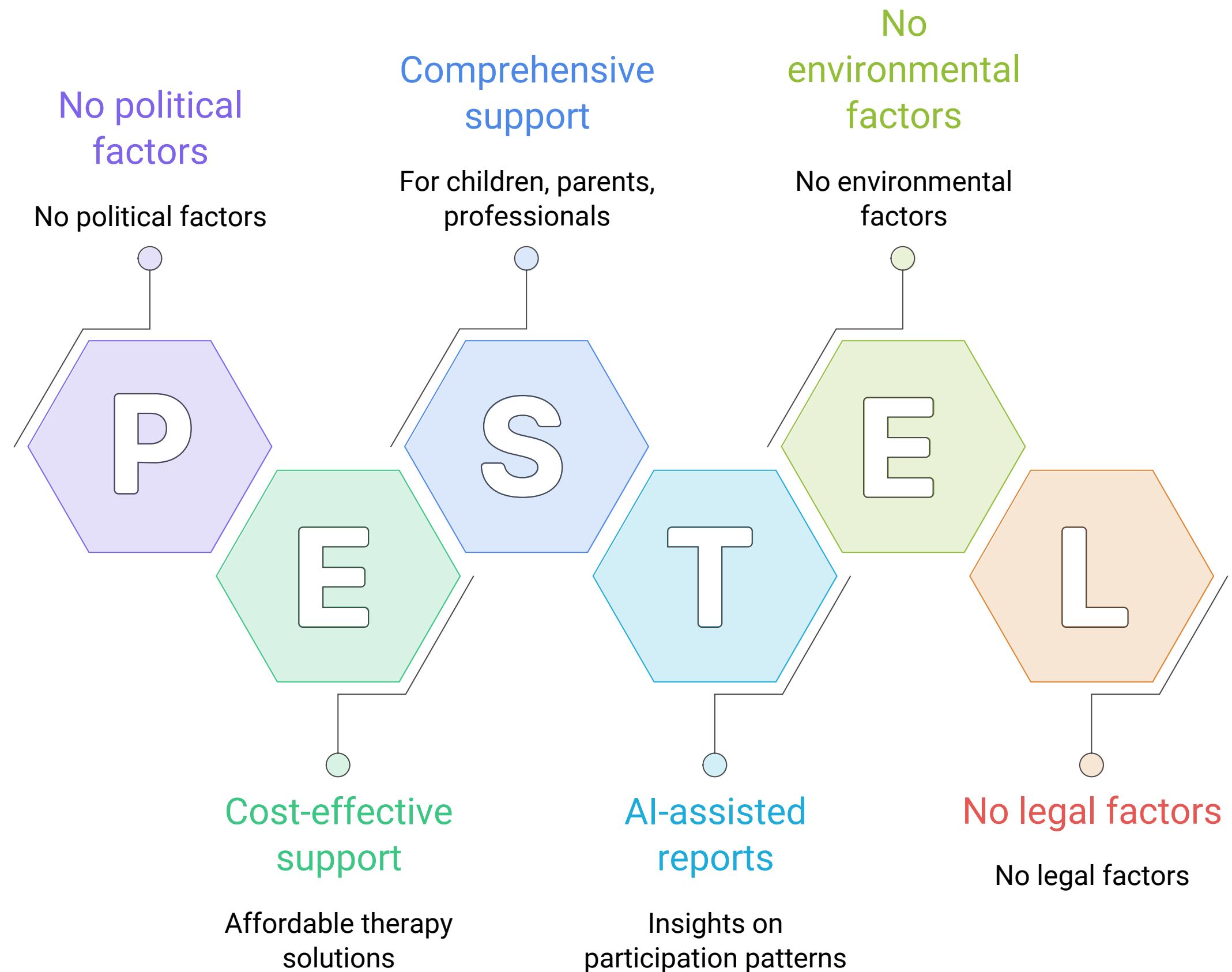
Productivity Loop



Key Platform Features

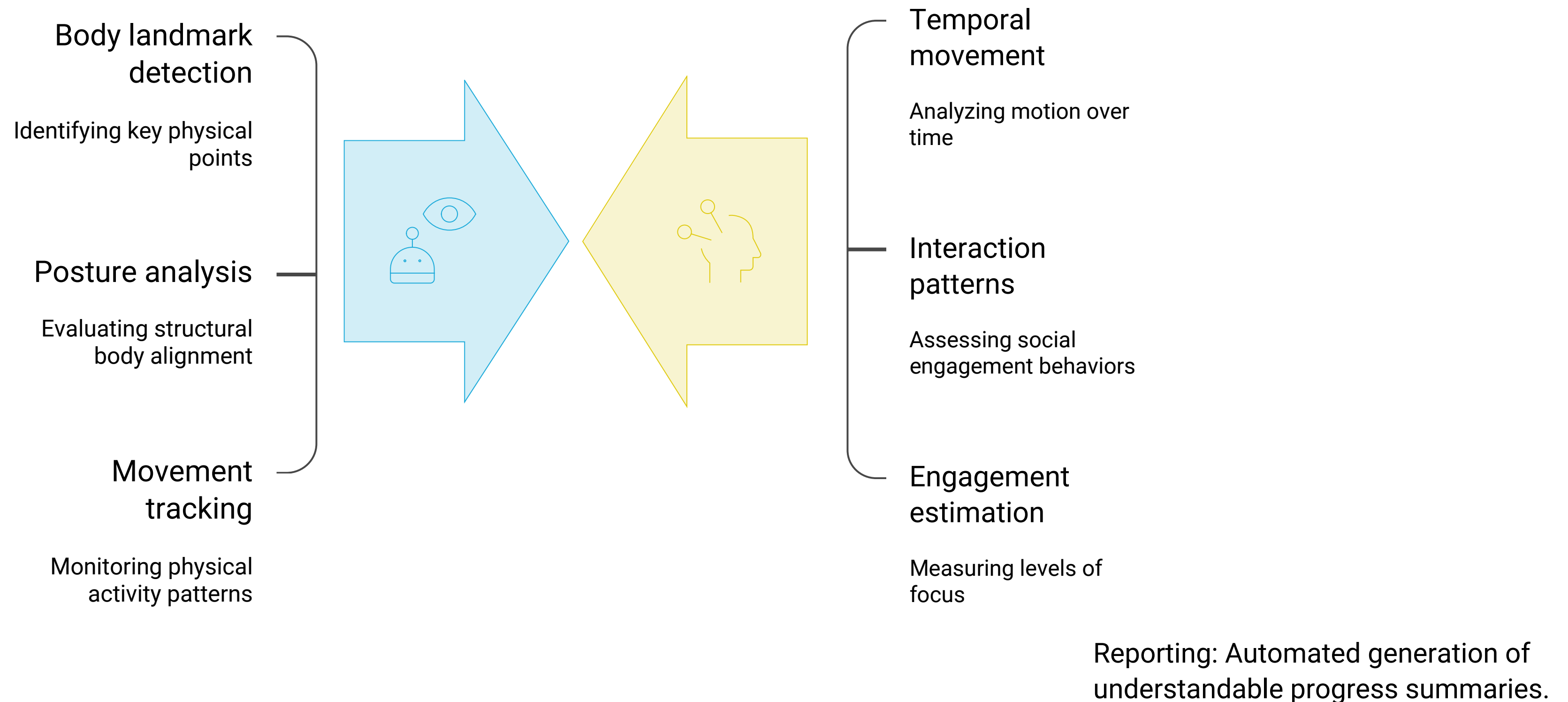
Comprehensive support for children, parents, and professionals.

- Therapy Videos: Guided sessions with interactive elements.
- Exercise Section: Posture and repetition analysis.
- Interactive Games: Designed to maintain engagement.



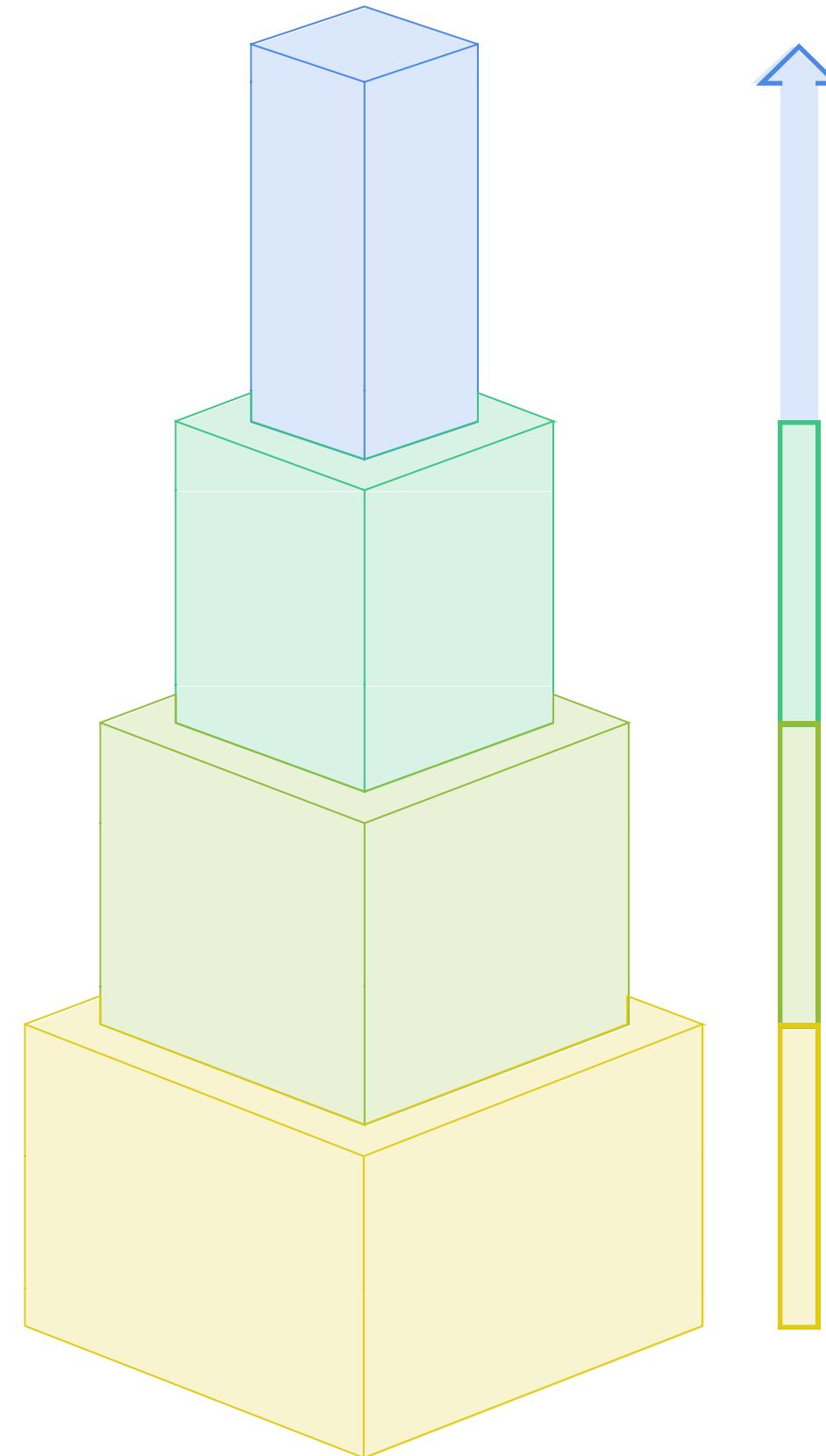
How It Works

AI provides supportive insights — it does not diagnose autism.



The Technology Stack

Built on a smartphone-first, modular architecture.



Mobile Development

Android Studio,
CameraX, Coroutines, Retrofit,
SplashUI

Backend Services

Python, FastAPI,
PostgreSQL

AI/CV Integration

CNN, MediaPipe, ML Kit, MedGemma,
NumPy, Torch, Pandas, SeaBorn, Matplotlib

Deployment

Docker
, Github

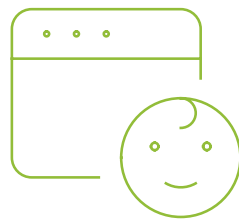
Feasibility, Limitations & Impact

Feasibility: Smartphone-first, modular MVP approach.

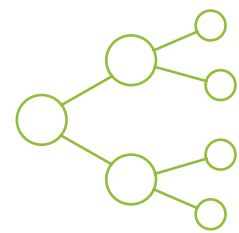
Pros



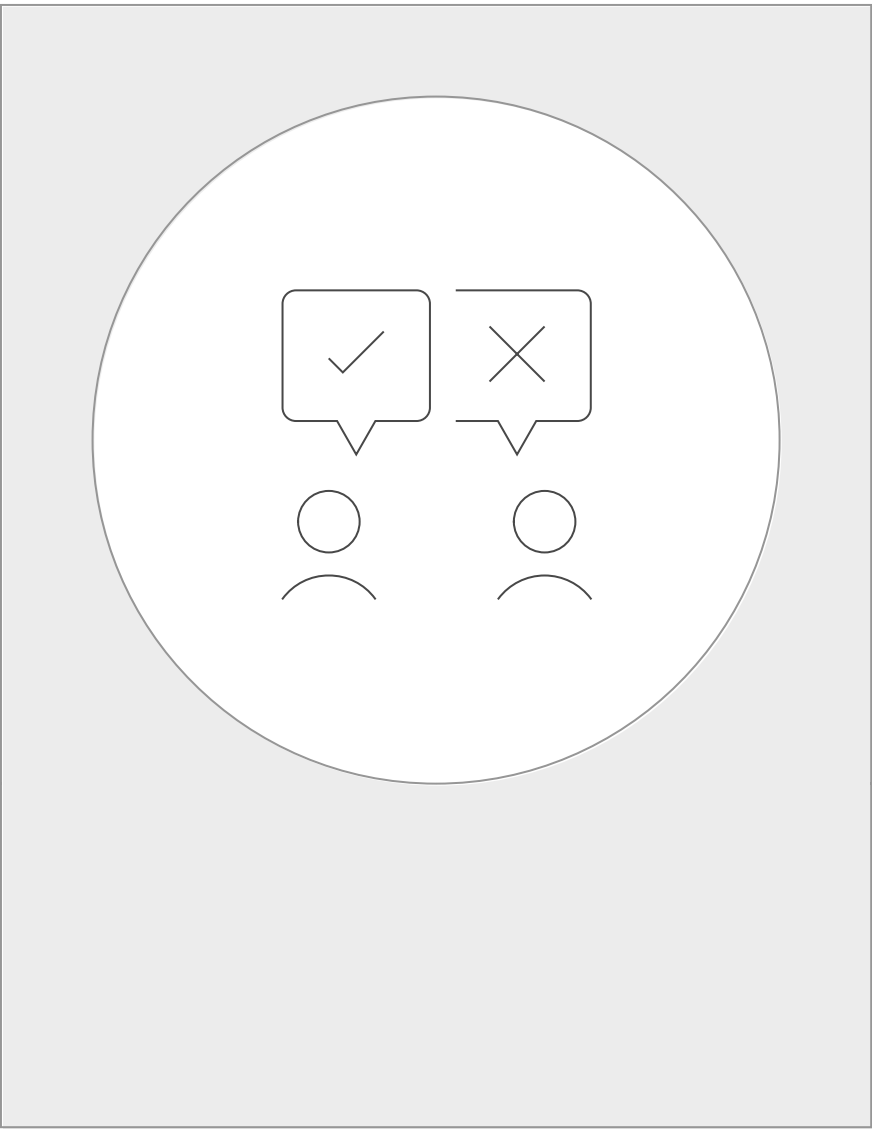
Personalized activities



Parent visibility

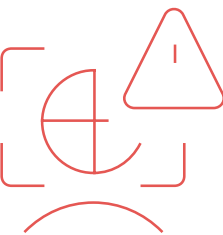


Scalable support



Cons

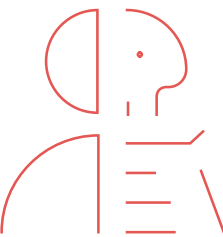
Camera issues



Continuous engagement



AI limitations

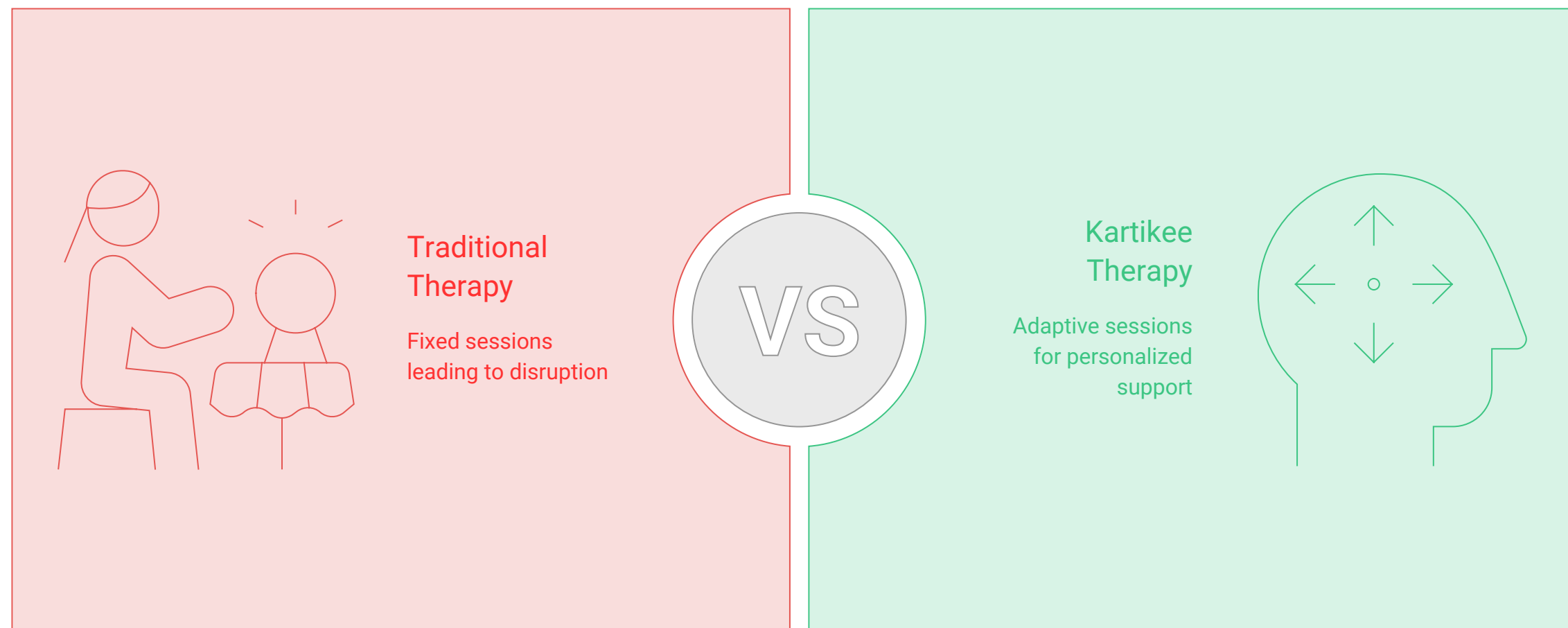


The Core Idea

What if therapy could adapt to the child instead of expecting every child to adapt to the therapy?

Kartikee process:

Therapy → Observe → Analyze → Adapt → Resume → Report.



Thank You